ROAR

Bringing the Pulse of the Stadium to Every Panther

Senior Capstone Project | FIU SCIS

The Problem

College life is a delicate balance. Between heavy coursework, part-time jobs, and research labs, students often miss the defining moments of campus life.

- Busy schedules prevent game attendance
- Generic social media is too cluttered
- FOMO (Fear of Missing Out) impact on student morale
- Fragmented sports news sources

The Motivation

Panther Pride

Creating a dedicated digital stadium for FIU's 50,000+ students and alumni to stay connected.

Native Experience

Leveraging high-performance iOS capabilities to deliver a fluid, TikTok-style engagement model.

THE SOLUTION

A niche social platform designed for zero-noise sports highlights.

iOS Exclusive

30s Limit

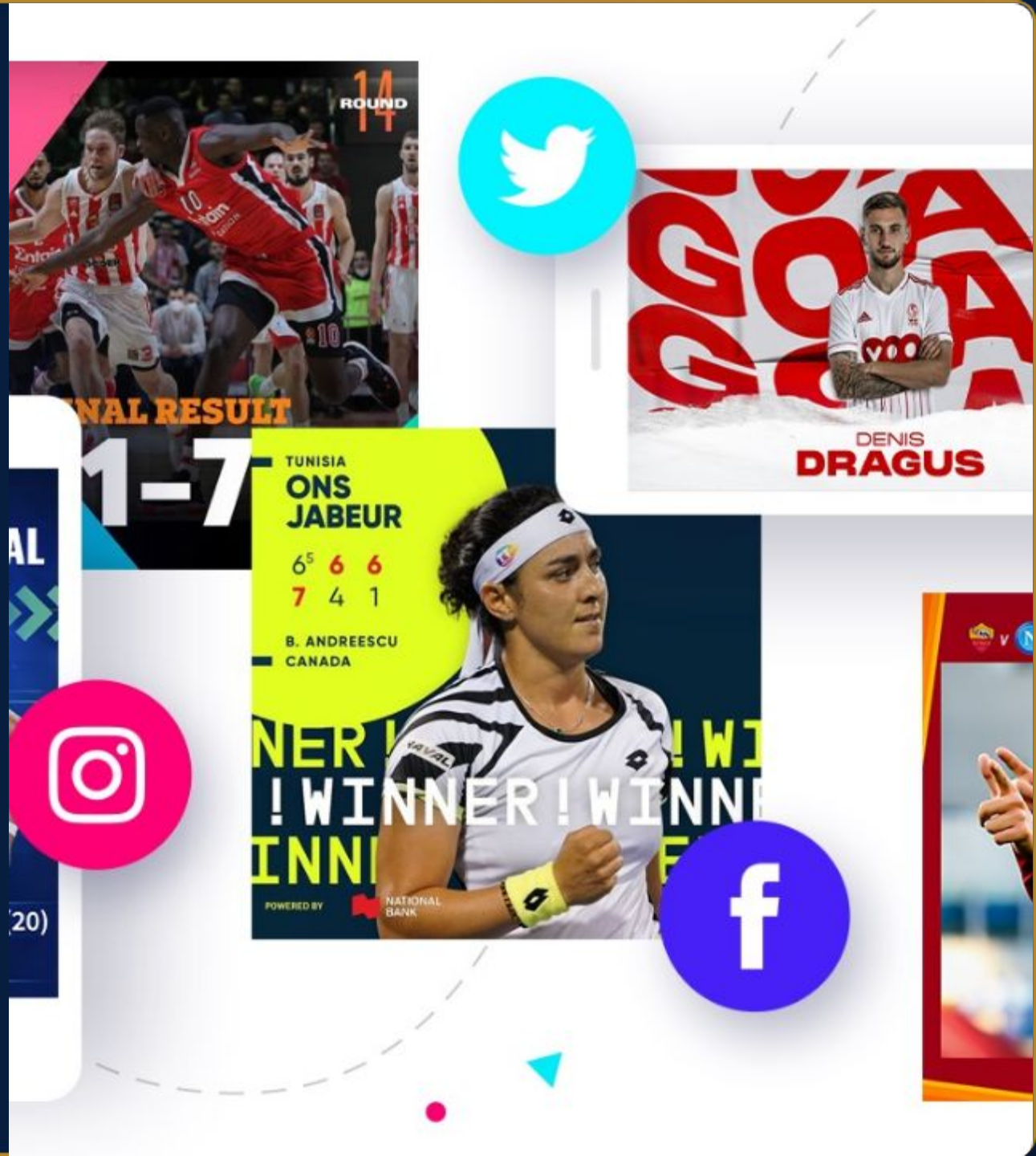
AWS Powered

UI/UX Strategy

Branding: Deep Blue & Radiant Gold representing FIU's identity.

Interface: Vertical scroll feed with 100% immersive video playback.

Accessibility: Full VoiceOver support and dynamic text scaling for an inclusive community experience.



System Architecture

Serverless Backend

- **S3:** High-speed video binary storage
- **Cognito:** Secure user authentication
- **Node.js** API
- **PostgreSQL:** Video duration & metadata validation

Data Modeling

Table (DynamoDB)	Partition Key	Sort Key	Purpose
Posts	postID	timestamp	Main video feed and tags
Users	userID	handle	Profile data and verification
Huddles	teamID	messageID	Real-time group chat logs
Engagement	postID	userID	Likes, views, and viral weight

Performance Metrics

Optimization via Caching

We implemented predictive pre-fetching and local video caching to ensure a zero-lag experience.

300ms average time-to-play for top videos in the "For You" feed.

Community Features

Huddles

Real-time chat rooms for every FIU team.

Live Polls

Interactive voting during live game windows.

Moderation

Admin tools to keep content safe and clean.

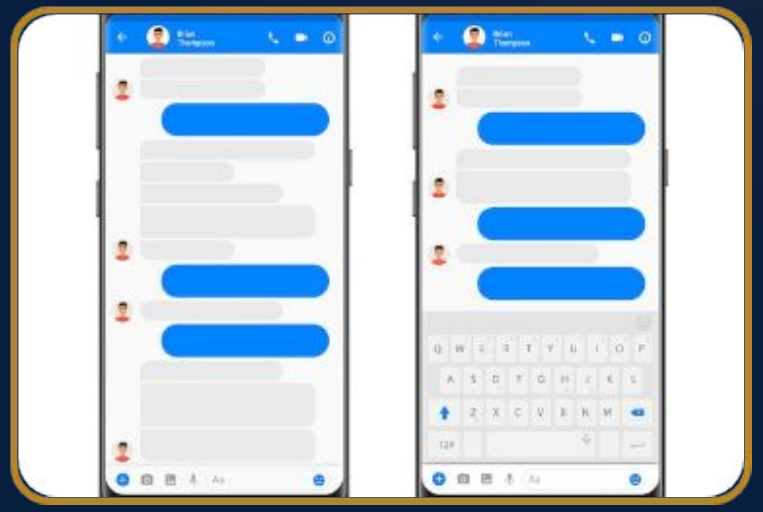
The Development Team

Xavier Williams

Project Lead

Alejandro & Richard

Backend & Cloud



Jordan & Zerong

iOS & UX Design

Thank You

Questions? Let's Roar.

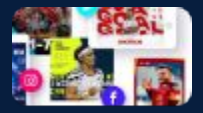
github.com/fiu-roar

roar-app.fiu.edu

Image Sources

https://ed.stanford.edu/sites/ed/files/news_images/overloadedmain_1.jpg

Source: ed.stanford.edu



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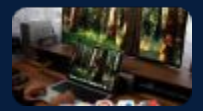

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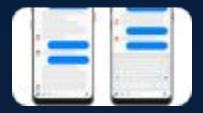
Source: www.youtube.com



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Source: www.freepik.com

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